

# Attach cloud storage

Instructions on how to attach cloud storage using UI

In CVAT you can use **AWS S3**, **Azure Blob Storage** and **Google Cloud Storage** storages to import and export image datasets for your tasks.

See:

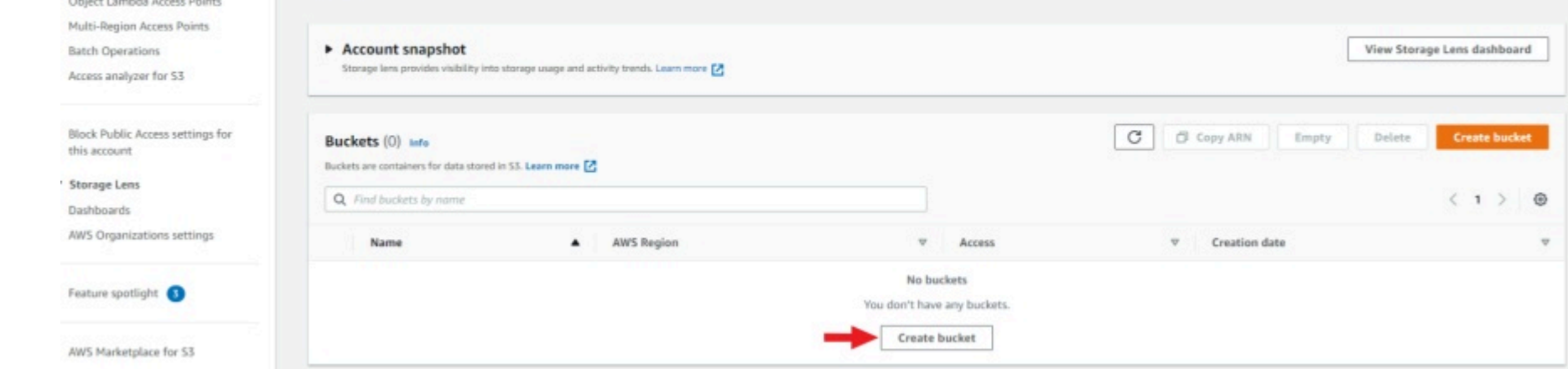
- [AWS S3](#)
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## AWS S3

### Create a bucket

To create bucket, do the following:

1. Create an [AWS account](#).
2. Go to [console AWS-S3](#), and click **Create bucket**.



3. Specify the name and region of the bucket. You can also copy the settings of another bucket by clicking on the **Choose bucket** button.

4. Enable **Block all public access**. For access, you will use **access key ID** and **secret access key**.

5. Click **Create bucket**.

A new bucket will appear on the list of buckets.

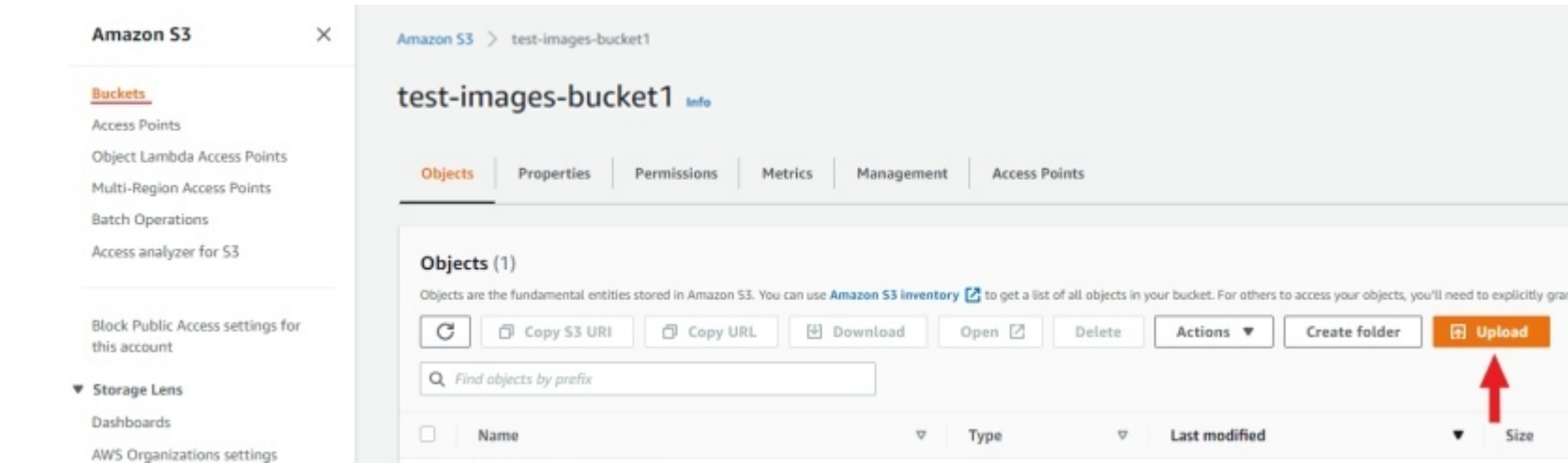
## Upload data

**Note:** manifest file is optional.

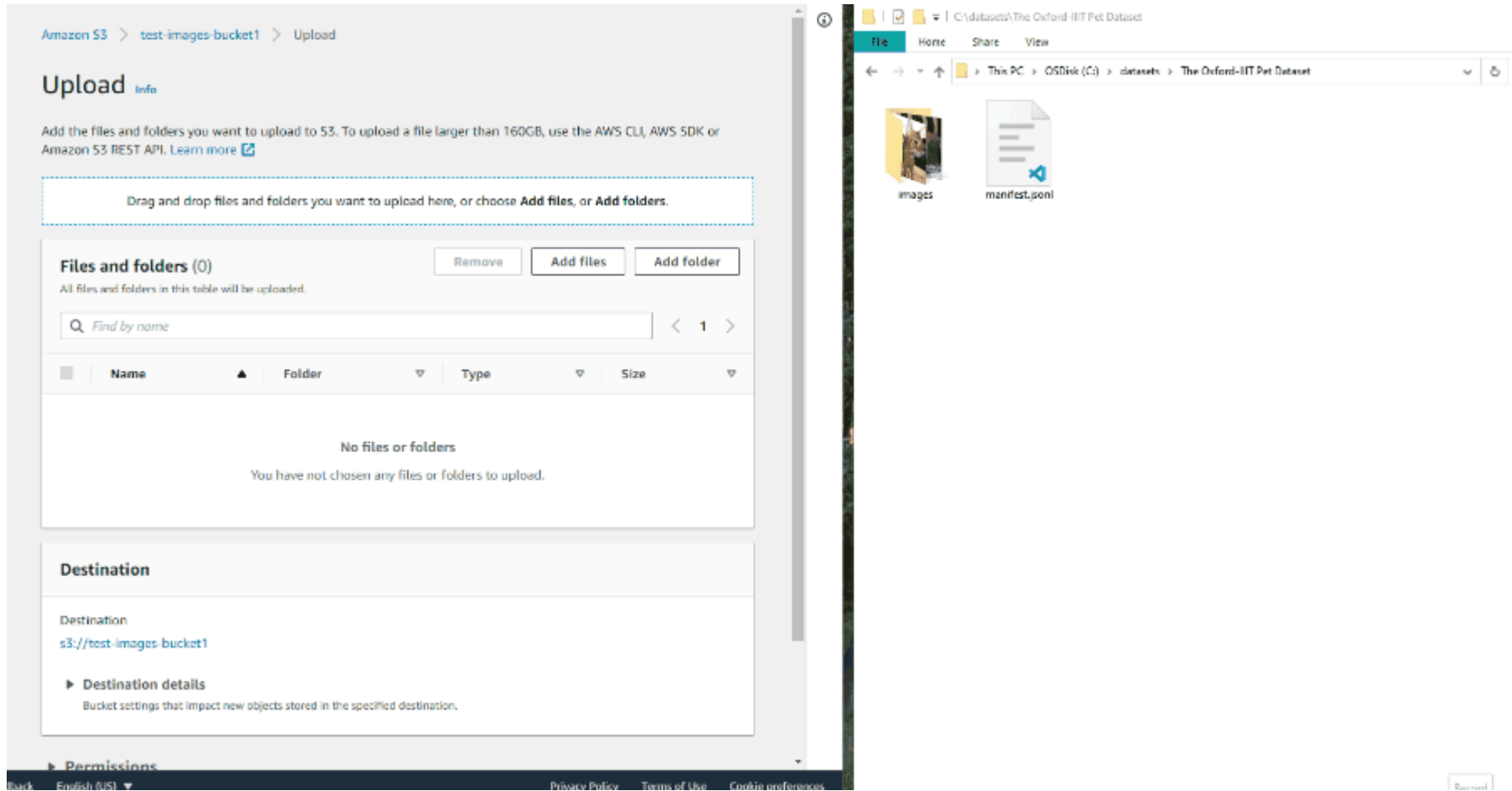
You need to upload data for annotation and the `manifest.jsonl` file.

1. Prepare data. For more information, see [prepare the dataset](#).

2. Open the bucket and click **Upload**.



3. Drag the manifest file and image folder on the page and click **Upload**:



## Access permissions

To add access permissions, do the following:

- 1. Go to [IAM](#) and click **Add users**.
- 2. Set **User name** and enable **Access key - programmatic access**.

Add user

1

2

3

4

5

Set user details

You can add multiple users at once with the same access type and permissions. [Learn more](#)

User name\*

user-1

⬇

Add another user

Select AWS access type

Select how these users will primarily access AWS. If you choose only programmatic access, it does NOT prevent users from accessing the console using an assumed role. Access keys and autogenerated passwords are provided in the last step. [Learn more](#)

Select AWS credential type\*

☒ Access key - Programmatic access

Enables an **access key ID** and **secret access key** for the AWS API, CLI, SDK, and other development tools.

☐ Password - AWS Management Console access

Enables a **password** that allows users to sign-in to the AWS Management Console.

\* Required

Cancel

Next: Permissions

- 3. Click **Next: Permissions**.
- 4. Click **Create group**, enter the group name.
- 5. Use search to find and select:
  - For read-only access: **AmazonS3ReadOnlyAccess**.
  - For full access: **AmazonS3FullAccess**.

Get permissions

Add user to group

Copy permissions from existing user

Attach existing policies directly

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Add user to group

Create group

Refresh

Search

Showing 1 result

| Group                                     | Attached policies |
|-------------------------------------------|-------------------|
| <input checked="" type="checkbox"/> users | None              |

Set permissions boundary

Set a permissions boundary to control the maximum permissions this user can have. This is an advanced feature used to delegate permission management to others. [Learn more](#)

☐ Create user without a permissions boundary

☒ Use a permissions boundary to control the maximum user permissions

Select policy to set the permissions boundary

Create policy

Refresh

Filter policies

Search s3

Showing 9 results

|                                  | Policy name                                             | Type        | Used as                |
|----------------------------------|---------------------------------------------------------|-------------|------------------------|
| <input type="radio"/>            | AmazonDMSRedshiftS3Role                                 | AWS managed | None                   |
| <input type="radio"/>            | AmazonS3FullAccess                                      | AWS managed | None                   |
| <input type="radio"/>            | AmazonS3ObjectLambdaExecutionRolePolicy                 | AWS managed | None                   |
| <input type="radio"/>            | AmazonS3OutpostsFullAccess                              | AWS managed | None                   |
| <input type="radio"/>            | AmazonS3OutpostsReadOnlyAccess                          | AWS managed | None                   |
| <input checked="" type="radio"/> | AmazonS3ReadOnlyAccess                                  | AWS managed | Permissions policy (1) |
| <input type="radio"/>            | IVSRecordToS3                                           | AWS managed | None                   |
| <input type="radio"/>            | QuickSightAccessForS3StorageManagementAnalyticsReadOnly | AWS managed | None                   |

Cancel

Previous

Next: Tags

6. (Optional) Add tags for the user and go to the next page.

7. Save **Access key ID** and **Secret access key**.

Add user

1

2

3

4

5

Success

You successfully created the users shown below. You can view and download user security credentials. You can also email users instructions for signing in to the AWS Management Console. This is the last time these credentials will be available to download. However, you can create new credentials at any time.

Users with AWS Management Console access can sign-in at: [https://\[redacted\].signin.aws.amazon.com/console](https://[redacted].signin.aws.amazon.com/console)

Download .csv

| User                                    | Access key ID                                   | Secret access key                               |
|-----------------------------------------|-------------------------------------------------|-------------------------------------------------|
| <div><div></div><div>user-1</div></div> | <div><div>[redacted]</div><div>Show</div></div> | <div><div>[redacted]</div><div>Show</div></div> |

For more information, see [Creating an IAM user in your AWS account](#)

Anonymous access

On how to grant public access to the bucket, see [Configuring block public access settings for your S3 buckets](#)

Attach AWS S3 storage

To attach storage, do the following:

Fill in the following fields:

| CVAT                | AWS S3                                                                                                                                                                                                                                                       |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display name        | Preferred display name for your storage.                                                                                                                                                                                                                     |
| Description         | (Optional) Add description of storage.                                                                                                                                                                                                                       |
| Provider            | From drop-down list select <b>AWS S3</b> .                                                                                                                                                                                                                   |
| Bucket name         | Name of the <a href="#">Bucket</a> .                                                                                                                                                                                                                         |
| Authentication type | Depends on the bucket setup: <ul style="list-style-type: none"><li><b>Key id and secret access key pair</b>: available on <a href="#">IAM</a>.</li><li><b>Anonymous access</b>: for anonymous access. Public access to the bucket must be enabled.</li></ul> |
| Region              | (Optional) Choose a region from the list or add a new one. For more information, see <a href="#">Available locations</a> .                                                                                                                                   |
| Prefix              | (Optional) Prefix is used to filter bucket content. By setting a default prefix, you ensure that only data from a specific folder in the cloud is used in CVAT. This will affect which files you see when creating a task with cloud data.                   |
| Manifests           | (Optional) Click + <b>Add manifest</b> and enter the name of the manifest file with an extension. For example: <code>manifest.jsonl</code> .                                                                                                                 |

After filling in all the fields, click **Submit**.

### AWS S3 manifest file

**Note:** manifest file is optional.

To prepare the manifest file, do the following:

- Go to [AWS CLI](#) and run [script for prepare manifest file](#).
- Perform the installation, following the [aws-shell manual](#),  
You can configure credentials by running `aws configure`.  
You will need to enter `Access Key ID` and `Secret Access Key` as well as the region.

```
aws configure
Access Key ID: <your Access Key ID>
Secret Access Key: <your Secret Access Key>
```

Copy

- Copy the content of the bucket to a folder on your computer:

```
aws s3 cp <s3://bucket-name> <yourfolder> --recursive
```

Copy

- After copying the files, you can create a manifest file as described in [prepare manifest file section](#):

```
python <cvat repository>/utils/dataset_manifest/create.py --output-dir <yourfolder> <yourfolder>
```

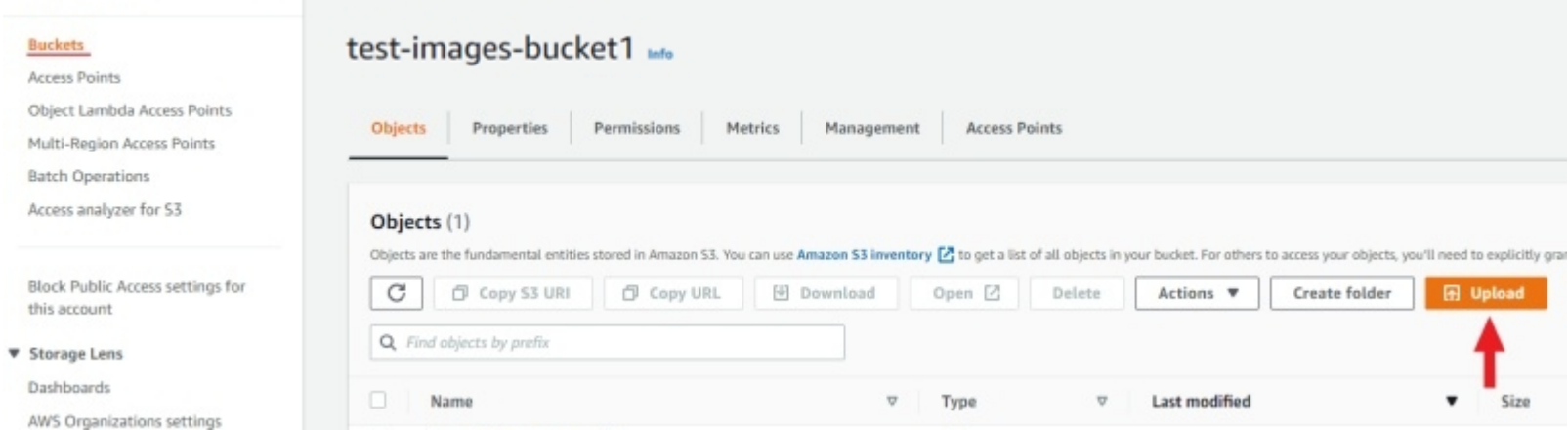
Copy

- When the manifest file is ready, upload it to aws s3 bucket:

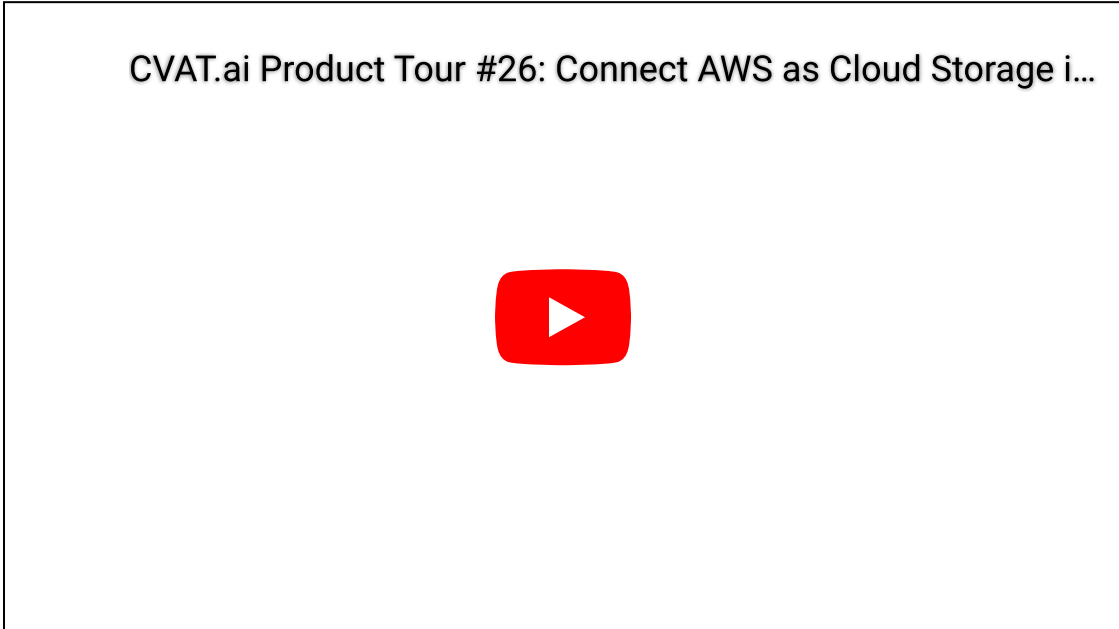
- For read and write permissions when you created the user, run:

```
aws s3 cp <yourfolder>/manifest.jsonl <s3://bucket-name>
```

Copy



Video tutorial: Add AWS S3 as Cloud Storage in CVAT



## Google Cloud Storage

### Create a bucket

To create bucket, do the following:

1. Create [Google account](#) and log into it.
2. On the [Google Cloud](#) page, click **Start Free**, then enter the required data and accept the terms of service.  
**Note:** Google requires to add payment, you will need a bank card to accomplish step 2.
3. [Create a Bucket](#) with the following parameters:
  - **Name your bucket:** Unique name.
  - **Choose where to store your data:** Set up a location nearest to you.
  - **Choose a storage class for your data:** Set a default class > Standard .
  - **Choose how to control access to objects:** Enforce public access prevention on this bucket > Uniform (default).
  - **How to protect data:** None

cvat-demo-bucket-02

Tip: Don't include any sensitive information

LABELS (OPTIONAL)

CONTINUE

Choose where to store your data

This choice defines the geographic placement of your data and affects cos performance, and availability. Cannot be changed later. [Learn more](#)

Location type

Multi-region

Highest availability across largest area

us (multiple regions in United States)

Dual-region

High availability and low latency across 2 regions

Region

Lowest latency within a single region

CONTINUE

Choose how to control access to objects

Prevent public access

Restrict data from being publicly accessible via the internet. Will prevent this bucket from being used for web hosting. [Learn more](#)

☒ Enforce public access prevention on this bucket

Access control

Uniform

Ensure uniform access to all objects in the bucket by using only bucket-level permissions (IAM). This option becomes permanent after 90 days. [Learn more](#)

Fine-grained

Specify access to individual objects by using object-level permissions (ACLs) in addition to your bucket-level permissions (IAM). [Learn more](#)

CONTINUE

Choose how to protect object data

Your data is always protected with Cloud Storage but you can also choose from these additional data protection options to prevent data loss. Note that object versioning and retention policies cannot be used together.

Protection tools

None

Object versioning (best for data recovery)

For restoring deleted or overwritten objects. To minimize the cost of storing versions, we recommend limiting the number of noncurrent versions per object and scheduling them to expire after a number of days. [Learn more](#)

Retention policy (best for compliance)

For preventing the deletion or modification of the bucket's objects for a specified minimum duration of time after being uploaded. [Learn more](#)

CONTINUE

You will be forwarded to the bucket.

## Upload data

**Note:** manifest file is optional.

You need to upload data for annotation and the `manifest.jsonl` file.

1. Prepare data. For more information, see [prepare the dataset](#).
2. Open the bucket and from the top menu select **Upload files** or **Upload folder** (depends on how your files are organized).

## Access permissions

To access Google Cloud Storage get a **Project ID** from [cloud resource manager page](#)

| Google Cloud Platform      |                  |                       |                  |        |         |        |      |
|----------------------------|------------------|-----------------------|------------------|--------|---------|--------|------|
| Manage resources           |                  |                       |                  |        |         |        |      |
| Filter                     |                  |                       |                  |        |         |        |      |
| <input type="checkbox"/>   | Name             | ID                    | Last accessed    | Status | Charges | Labels | Tags |
| <input type="checkbox"/>   | No organization  |                       | December 6, 2021 |        |         |        |      |
| <input type="checkbox"/>   | My First Project | project-xxxxxx-xxxxxx | December 6, 2021 |        |         |        |      |
| RESOURCES PENDING DELETION |                  |                       |                  |        |         |        |      |

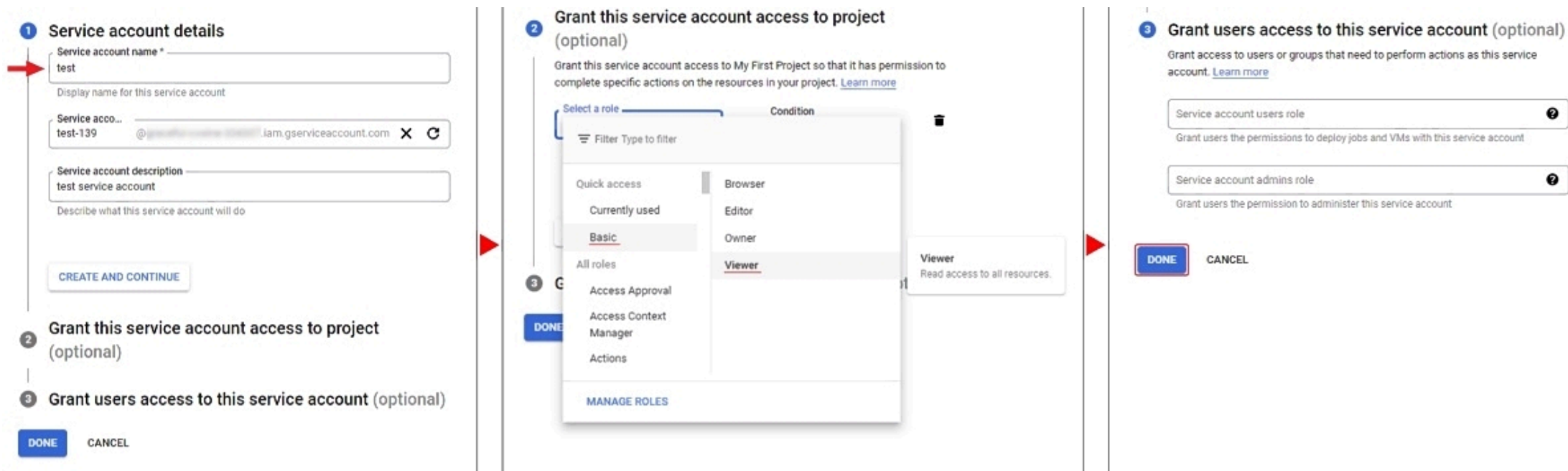
And follow instructions below based on the preferable type of access.

### Authenticated access

For authenticated access you need to create a service account and key file.

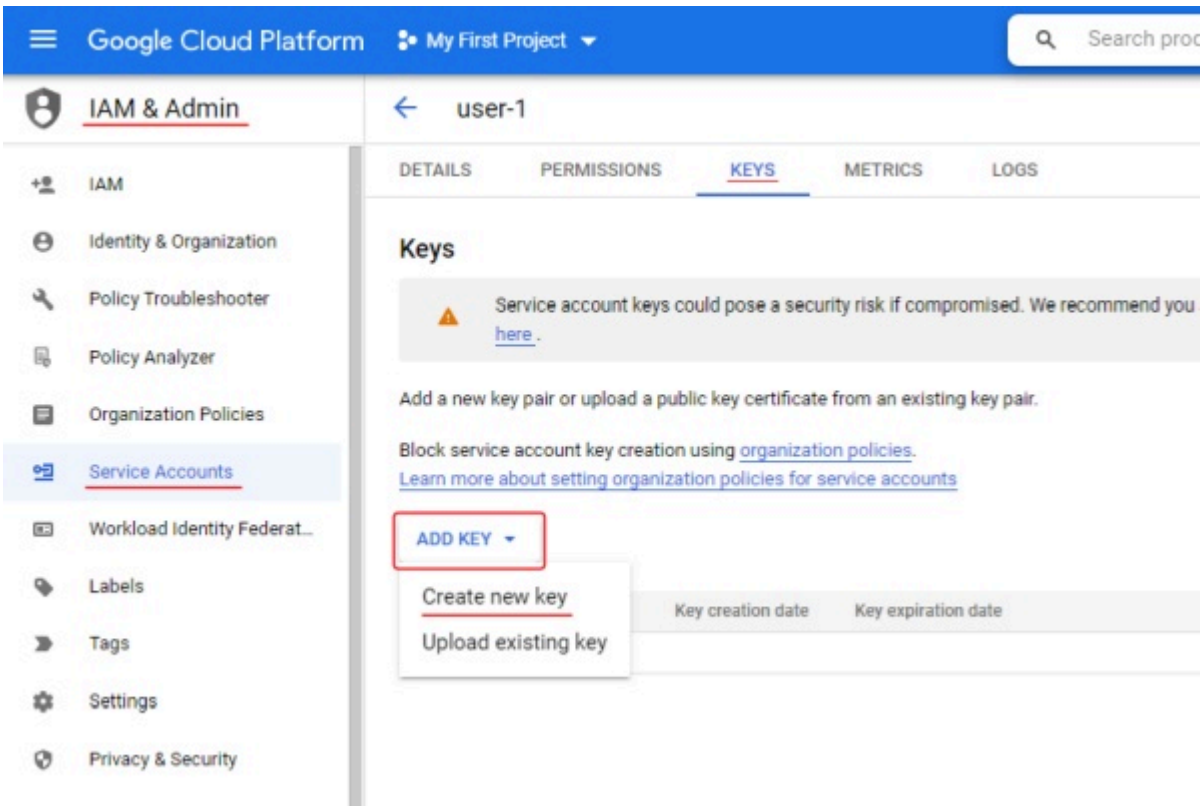
To create a service account:

- 3. Select a role, for example **Basic > Viewer**, and click **Continue**.
- 4. (Optional) Give access rights to the service account.
- 5. Click **Done**.



To create a key:

- 1. Go to **IAM & Admin > Service Accounts** > click on account name > **Keys**.
- 2. Click **Add key** and select **Create new key > JSON**
- 3. Click **Create**. The key file will be downloaded automatically.

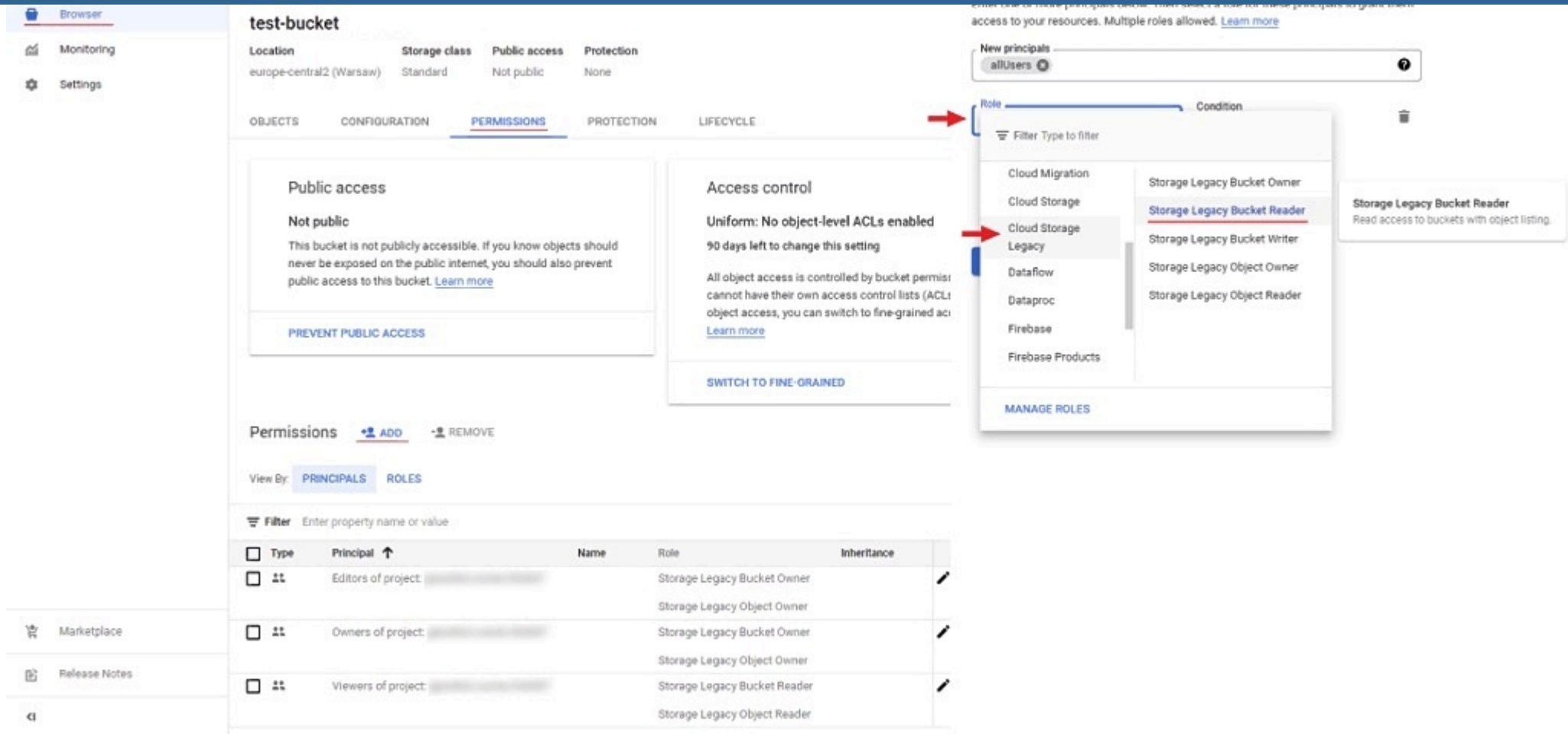


For more information about keys, see [Learn more about creating keys](#).

Anonymous access

To configure anonymous access:

- 1. Open the bucket and go to the **Permissions** tab.
- 2. Click **+ Grant access** to add new principals.
- 3. In the **New principals** field specify `allUsers` , select roles: `Cloud Storage Legacy > Storage Legacy Bucket Reader` .
- 4. Click **Save**.



Now you can attach the Google Cloud Storage bucket to CVAT.

## Attach Google Cloud Storage

To attach storage, do the following:

1. Log into CVAT and in the separate tab open your [bucket](#) page.
2. In the CVAT, on the top menu select **Cloud storages** > on the opened page click **+**.

Fill in the following fields:

| CVAT                | Google Cloud Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display name        | Preferred display name for your storage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Description         | (Optional) Add description of storage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Provider            | From drop-down list select <b>Google Cloud Storage</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Bucket name         | Name of the bucket. You can find it on the <a href="#">storage browser page</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Authentication type | Depends on the bucket setup: <ul style="list-style-type: none"><li>• <b>Authenticated access:</b> Click on the <b>Key file</b> field and upload key file from computer.</li><li><b>Advanced:</b> For self-hosted solution, if the key file was not attached, then environment variable <code>GOOGLE_APPLICATION_CREDENTIALS</code> that was specified for an environment will be used. For more information, see <a href="#">Authenticate to Cloud services using client libraries</a>.</li><li>• <b>Anonymous access:</b> for anonymous access. Public access to the bucket must be enabled.</li></ul> |
| Prefix              | (Optional) Used to filter data from the bucket. By setting a default prefix, you ensure that only data from a specific folder in the cloud is used in CVAT. This will affect which files you see when creating a task with cloud data.                                                                                                                                                                                                                                                                                                                                                                  |
| Project ID          | <a href="#">Project ID</a> .<br>For more information, see <a href="#">projects page</a> and <a href="#">cloud resource manager page</a> .<br><b>Note:</b> Project name does not match the project ID.                                                                                                                                                                                                                                                                                                                                                                                                   |
| Location            | (Optional) Choose a region from the list or add a new one. For more information, see <a href="#">Available locations</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Manifests           | (Optional) Click <b>+ Add manifest</b> and enter the name of the manifest file with an extension. For example: <code>manifest.jsonl</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

After filling in all the fields, click **Submit**.

## Video tutorial: Add Google Cloud Storage as Cloud Storage in CVAT

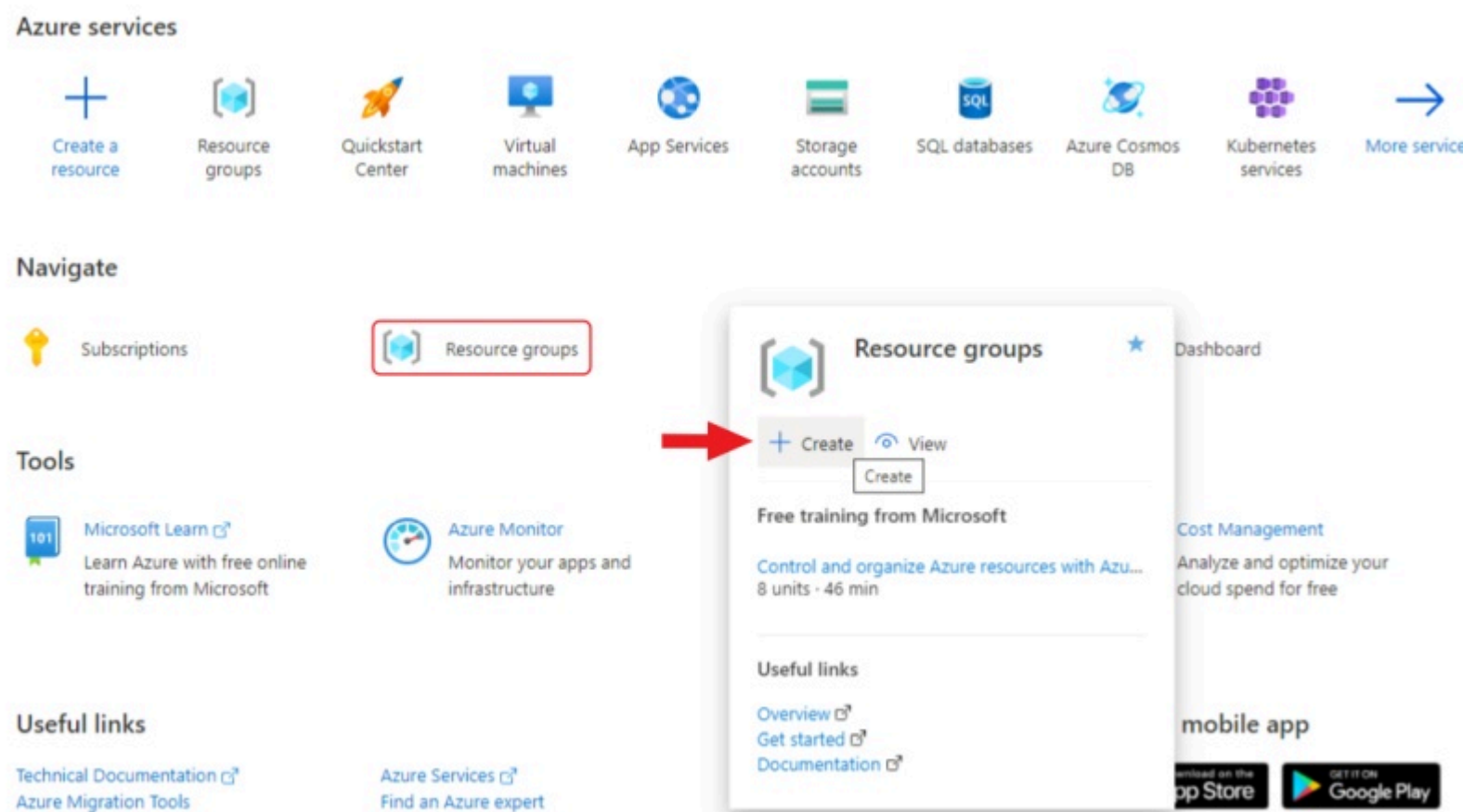


# Microsoft Azure Blob Storage

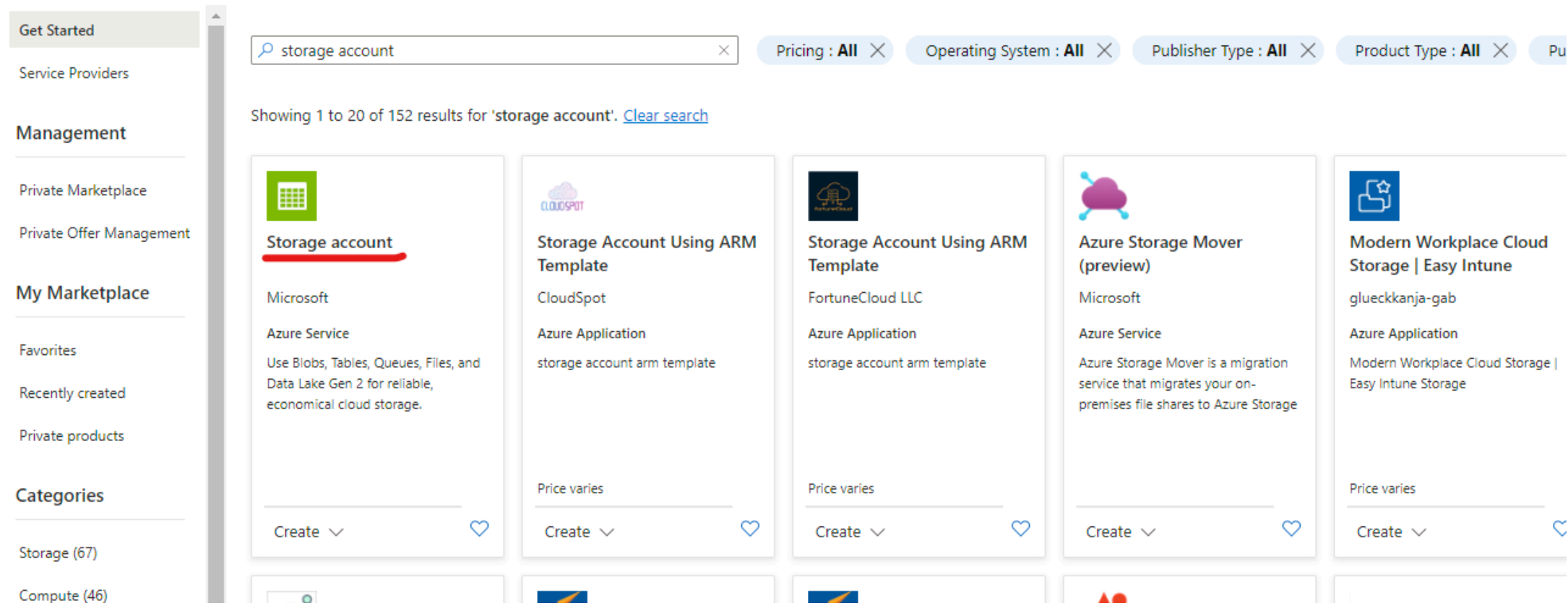
## Create a bucket

To create bucket, do the following:

- 1. Create an [Microsoft Azure](#) account and log into it.
- 2. Go to [Azure portal](#), hover over the resource , and in the pop-up window click **Create**.



- 3. Enter a name for the group and click **Review + create**, check the entered data and click **Create**.
- 4. Go to the [resource groups page](#), navigate to the group that you created and click **Create resources**.
- 5. On the marketplace page, use search to find **Storage account**.



6. Click on **Storage account** and on the next page click **Create**.

7. On the **Basics** tab, fill in the following fields:
- **Storage account name:** to access container from CVAT.
  - Select a region closest to you.
  - Select **Performance > Standard**.
  - Select **Local-redundancy storage (LRS)**.
  - Click **next: Advanced>**.

8. On the **Advanced** page, fill in the following fields:
- (Optional) Disable **Allow enabling public access on containers** to prohibit anonymous access to the container.
  - Click **Next > Networking**.

Network connectivity

You can connect to your storage account either publicly, via public IP addresses or service endpoints, or privately, using a private endpoint.

Network access \*

☒

Enable public access from all networks

☐

Enable public access from selected virtual networks and IP addresses

☐

Disable public access and use private access

Enabling public access from all networks might make this resource available publicly. Unless public access is required, we recommend using a more restricted access type. [Learn more](#)

Network routing

Determine how to route your traffic as it travels from the source to its Azure endpoint. Microsoft network routing is recommended for most customers.

Routing preference ⓘ \*

☒

Microsoft network routing

☐

Internet routing

9. On the **Networking** tab, fill in the following fields:

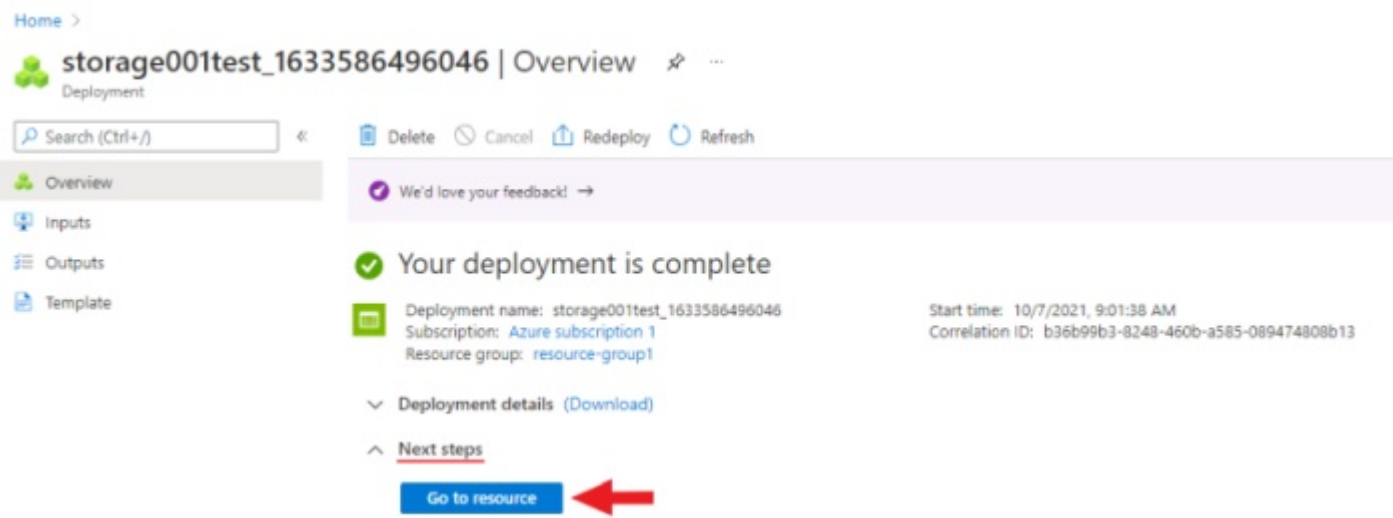
- If you want to change public access, enable **Public access from all networks**.
- Click **Next>Data protection**.

 You do not need to change anything in other tabs until you need some specific setup.

10. Click **Review** and wait for the data to load.

11. Click **Create**. Deployment will start.

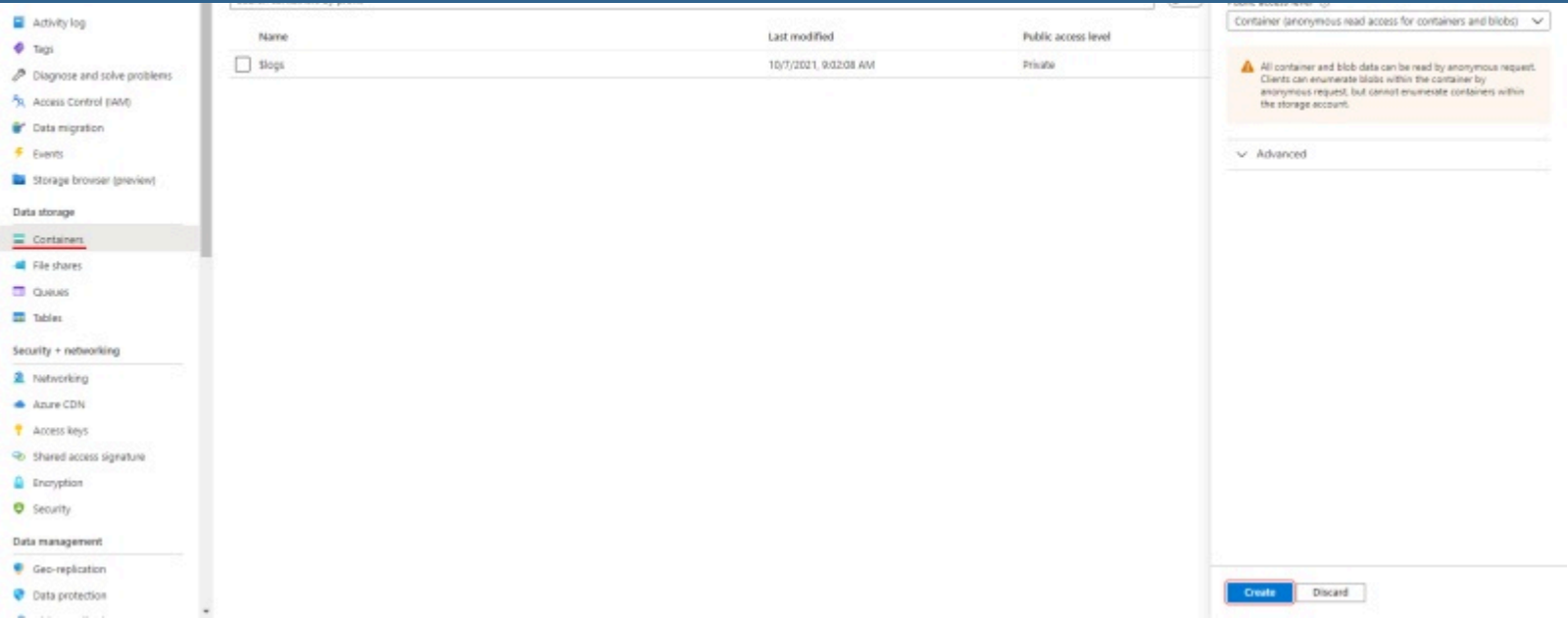
12. After deployment is over, click **Go to resource**.



Create a container

To create container, do the following:

1. Go to the containers section and on the top menu click **+Container**

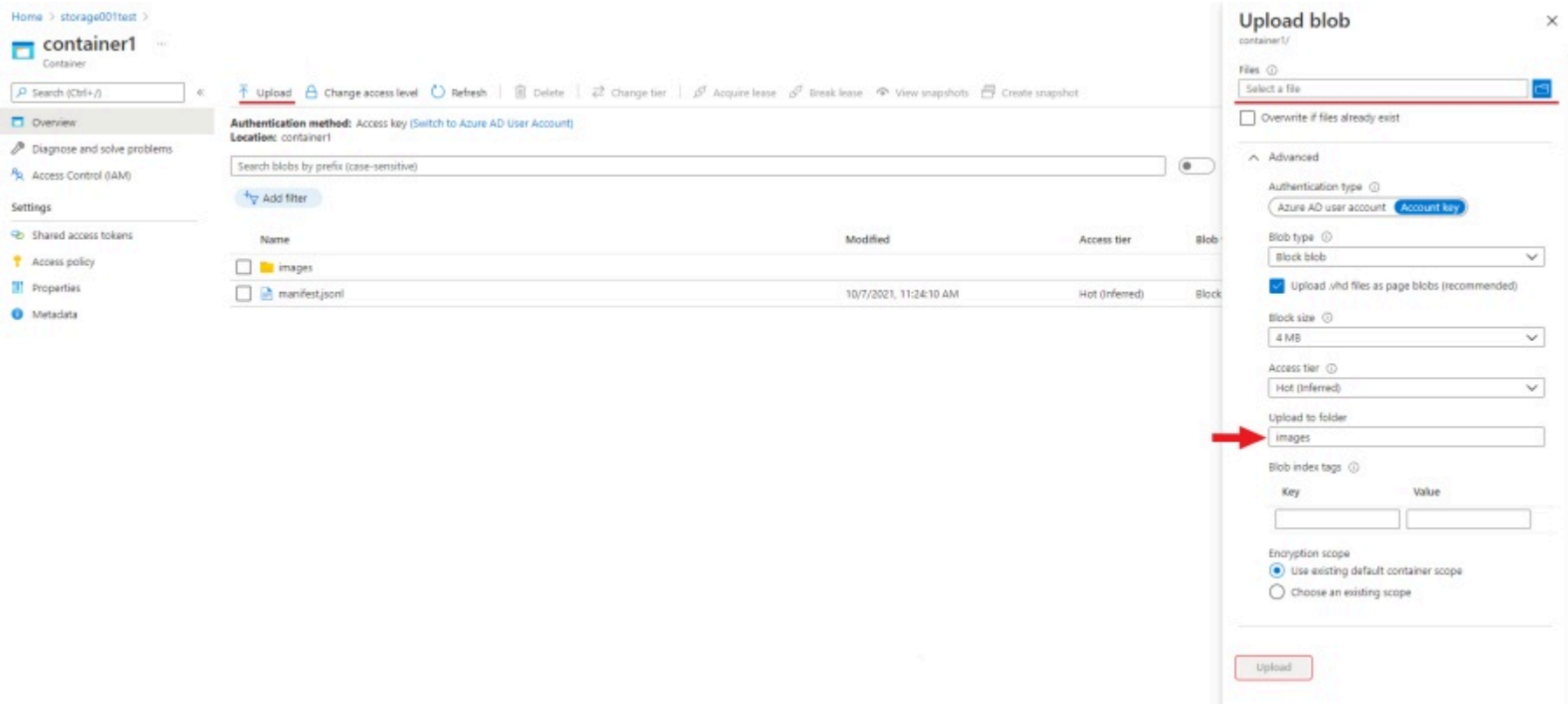


3. Enter the name of the container.
4. (Optional) In the **Public access level** drop-down, select type of the access.  
**Note:** this field will inactive if you disabled **Allow enabling public access on containers**.
5. Click **Create**.

## Upload data

You need to upload data for annotation and the `manifest.jsonl` file.

1. Prepare data. For more information, see [prepare the dataset](#).
2. Go to container and click **Upload**.
3. Click **Browse for files** and select images.  
 Note: If images are in folder, specify folder in the **Advanced settings > Upload to folder**.
4. Click **Upload**.



## SAS token and connection string

Use the SAS token or connection string to grant secure access to the container.

To configure the credentials:

1. Go to **Home > Resource groups > You resource name > Your storage account**.
2. On the left menu, click **Shared access signature**.
3. Change the following fields:
  - **Allowed services:** Enable **Blob** . Disable all other fields.
  - **Allowed resource types:** Enable **Container** and **Object**. Disable all other fields.
  - **Allowed permissions:** Enable **Read**, **Write**, and **List**. Disable all other fields.

- Leave all other fields with default parameters.

4. Click **Generate SAS and connection string** and copy **SAS token** or **Connection string**.

Home > storage001test

storage001test | Shared access signature

A shared access signature (SAS) is a URI that grants restricted access rights to Azure Storage resources. You can provide a shared access signature to clients who should not be trusted with your storage account key but whom you wish to delegate access to certain storage account resources. By distributing a shared access signature URI to these clients, you grant them access to a resource for a specified period of time.

An account-level SAS can delegate access to multiple storage services (i.e. blob, file, queue, table). Note that stored access policies are currently not supported for an account-level SAS.

Learn more

Allowed services ☐ Blob ☐ File ☐ Queue ☐ Table

Allowed resource types ☐ Service ☒ Container ☒ Object

Allowed permissions ☒ Read ☐ Write ☐ Delete ☒ List ☐ Add ☐ Create ☐ Update ☐ Process ☐ Immutable storage

Blob versioning permissions ☒ Enables deletion of versions

Allowed blob index permissions ☒ Read/Write ☒ Filter

Start and expiry date/time

Start 10/11/2021 11:43:54 AM

End 10/11/2021 7:43:54 PM

(UTC+03:00) --- Current Time Zone ---

Allowed IP addresses

For example, 168.1.5.65 or 168.1.5.65-168.1.5.70

Allowed protocols ☐ HTTPS only ☒ HTTPS and HTTP

Preferred routing tier ☒ Basic (default) ☐ Microsoft network routing ☐ Internet routing

Some routing options are disabled because the endpoints are not published.

Signing key

Generate SAS and connection string

Connection string

SAS token

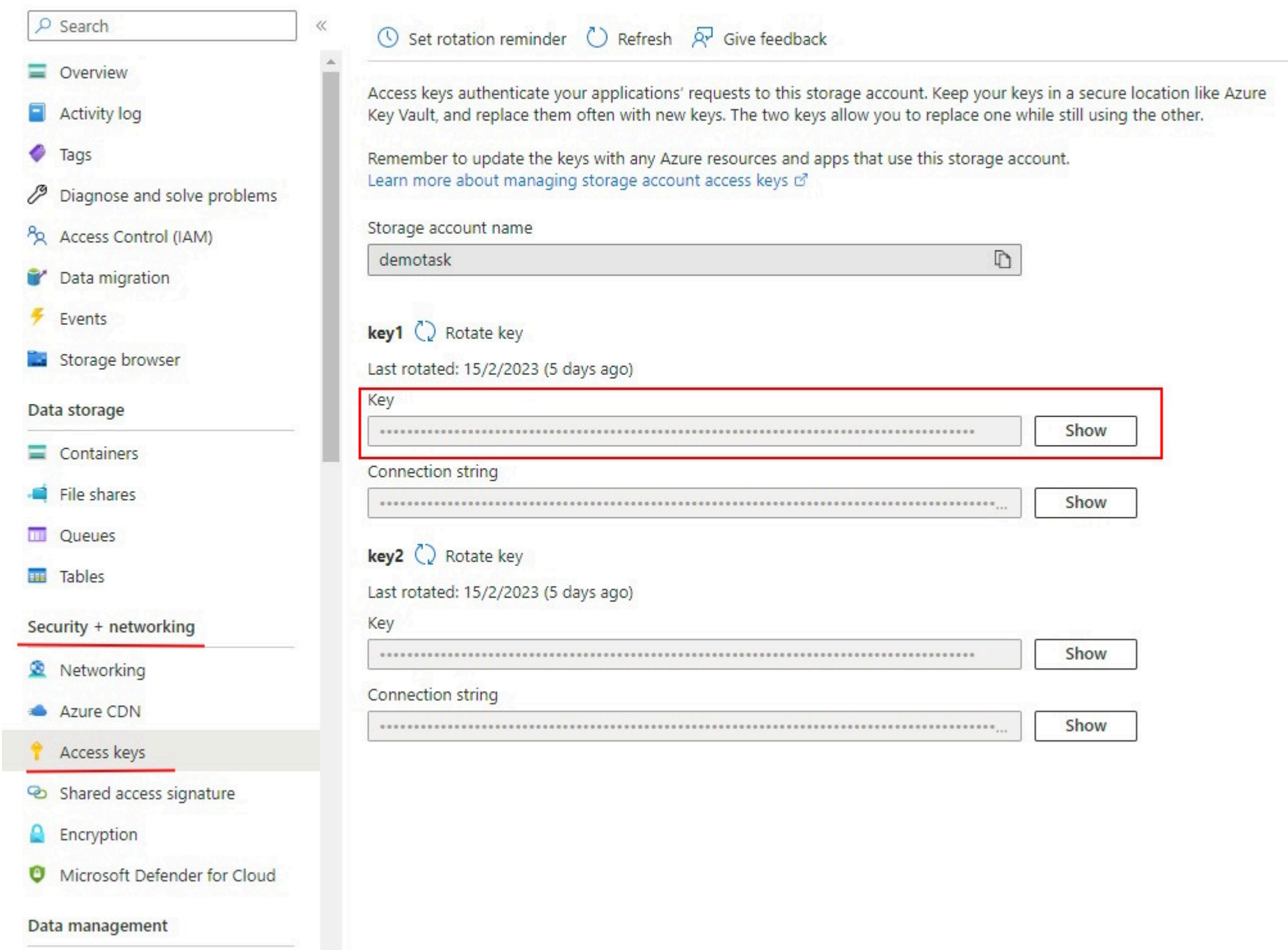
Blob service SAS URL

## Personal use

For personal use, you can use the **Access Key** from your storage account in the CVAT **SAS Token** field.

To get the **Access Key**:

1. In the Azure Portal, go to the **Security + networking > Access Keys**
2. Click **Show** and copy the key.



Attach Azure Blob Storage

To attach storage, do the following:

- 1. Log into CVAT and in the separate tab open your bucket page.
- 2. In the CVAT, on the top menu select **Cloud storages** > on the opened page click +.

Fill in the following fields:

| CVAT                | Azure                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display name        | Preferred display name for your storage.                                                                                                                                                                                                                                                                                                                                                                                   |
| Description         | (Optional) Add description of storage.                                                                                                                                                                                                                                                                                                                                                                                     |
| Provider            | From drop-down list select <b>Azure Blob Container</b> .                                                                                                                                                                                                                                                                                                                                                                   |
| Container name`     | Name of the cloud storage container.                                                                                                                                                                                                                                                                                                                                                                                       |
| Authentication type | Depends on the container setup.<br><b>Account name and SAS token:</b> <ul style="list-style-type: none"><li>• <b>Account name</b> enter storage account name.</li><li>• <b>SAS token</b> is located in the <b>Shared access signature</b> section of your <a href="#">Storage account</a>.</li></ul><br><b>. Anonymous access:</b> for anonymous access <b>Allow enabling public access on containers</b> must be enabled. |
| Prefix              | (Optional) Used to filter data from the bucket. By setting a default prefix, you ensure that only data from a specific folder in the cloud is used in CVAT. This will affect which files you see when creating a task with cloud data.                                                                                                                                                                                     |
| Manifests           | (Optional) Click + <b>Add manifest</b> and enter the name of the manifest file with an extension. For example: <code>manifest.jsonl</code> .                                                                                                                                                                                                                                                                               |

Video tutorial: Add Microsoft Azure Blob Storage as Cloud Storage in CVAT

CVAT.ai Product Tour #28: Connect Microsoft Azure as Clo...



# Prepare the dataset

For example, the dataset is [The Oxford-IIIT Pet Dataset](#):

- 1. Download the [archive with images](#).
- 2. Unpack the archive into the prepared folder.
- 3. Create a manifest. For more information, see [Dataset manifest](#):

```
python <cvat repository>/utils/dataset_manifest/create.py --output-dir <your_folder> <your_folder>
```

Copy